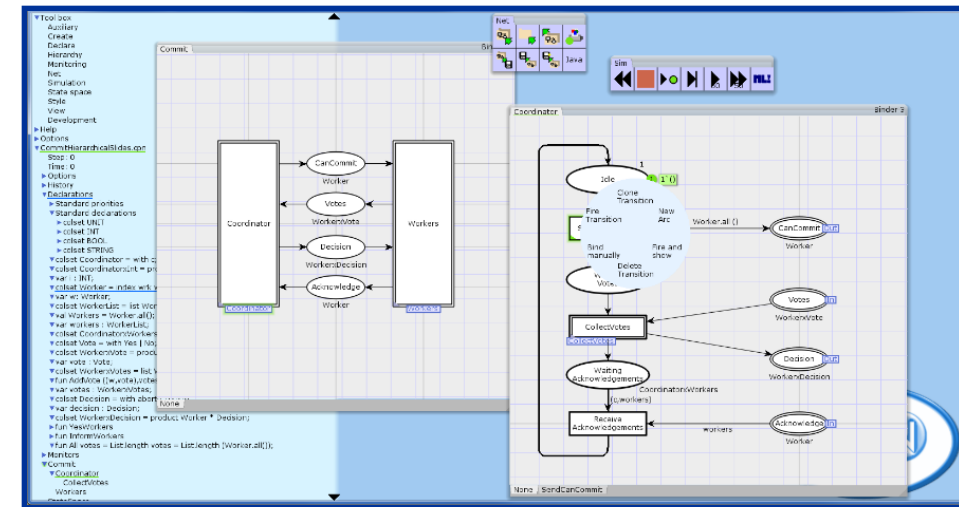
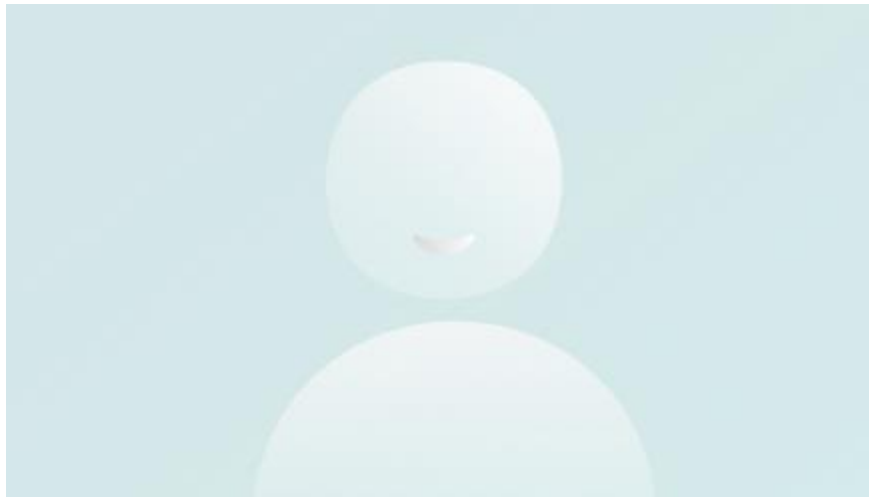


Motivation and overview of Coloured Petri Nets and CPN Tools



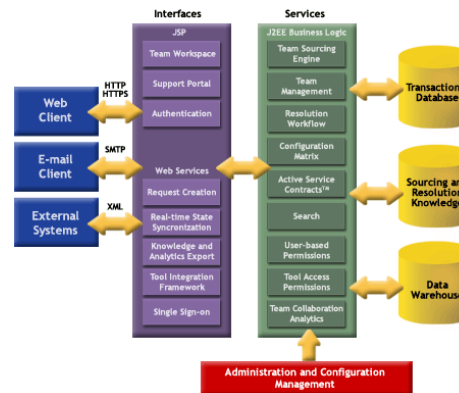
Lars Michael Kristensen
Department of Computer Science, Electrical Engineering, and Mathematical Sciences
Western Norway University of Applied Sciences

Concurrent and distributed systems

- The vast majority of system development projects today are concerned with **concurrent and distributed systems**
 - Structured as a collection of concurrently executing software components and applications (parallelism)
 - Operation relies on communication, synchronisation, and resource sharing



Internet protocols, cloud, IoT, web-based applications



Multi-core platforms and multi-threaded software



Automation systems and networked control systems

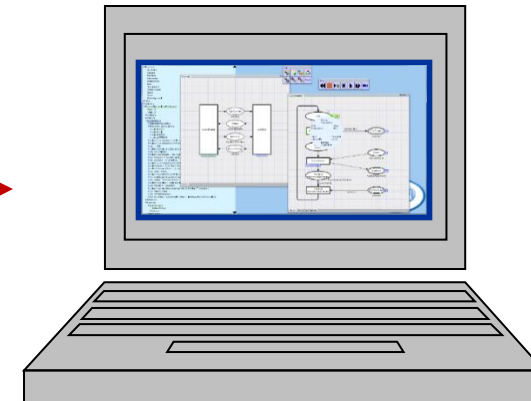
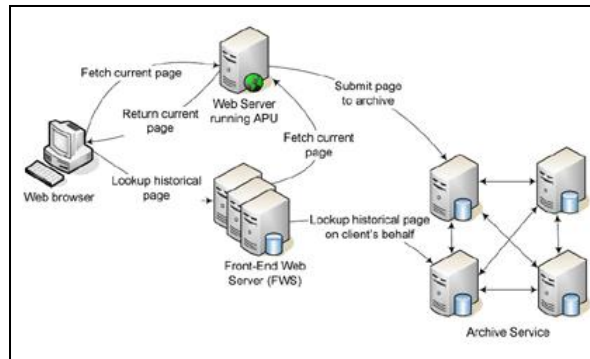
Complex behaviour

- **The engineering of concurrent and distributed systems is **challenging** due to their **complex behaviour****
 - Concurrently executing and independently scheduled components
 - Non-deterministic and asynchronous behaviour (e.g., timeouts, message loss, external events, ...)
 - Almost impossible for developers to have a complete understanding of the system behaviour
 - Testing is challenging and reproducing errors is often difficult
- **Methods to support the engineering of **reliable distributed and concurrent systems** are highly relevant**

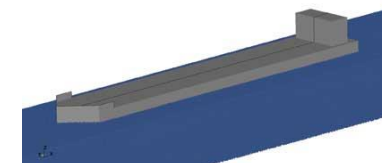


Modelling

- One way to approach these challenges is via the **construction of executable models**
- Models are **abstract representations** of a system which can be manipulated by software tools



- Modelling is widely used in most engineering disciplines

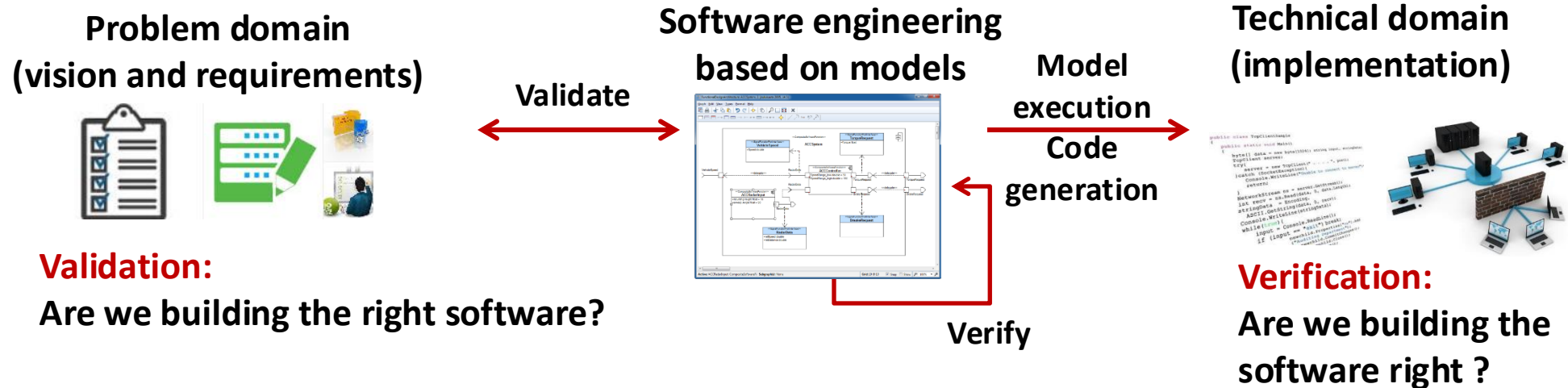


Why modelling?

- **Benefits of constructing executable models**
 - **Insight** into the design and operation of the system
 - **Completeness**: results in a more complete design
 - **Correctness**: reveal errors and ambiguities in early phases
- **Abstraction and communication** – early validation using high-level and domain-specific concepts in development
- **Reliability** – testing and verification prior to implementation and deployment
 - **Functional properties** (e.g., deadlocks, timing requirements,...)
 - **Performance properties** (e.g., delay, throughout, scalability,...)

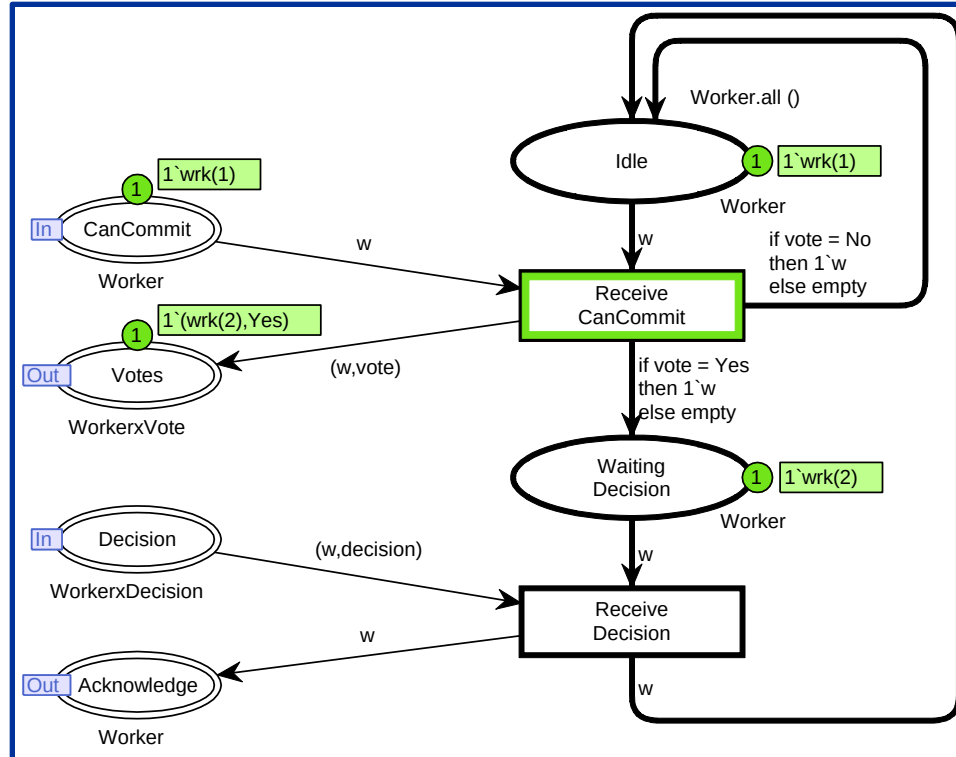
Model-driven engineering

- Models may in some cases also be used (directly or indirectly) as a basis for implementation



Coloured Petri Nets - CPNs

- General-purpose graphical modelling language for the engineering of **concurrent and distributed systems**
- Combines **Petri Nets** and a **programming language**



Petri Nets

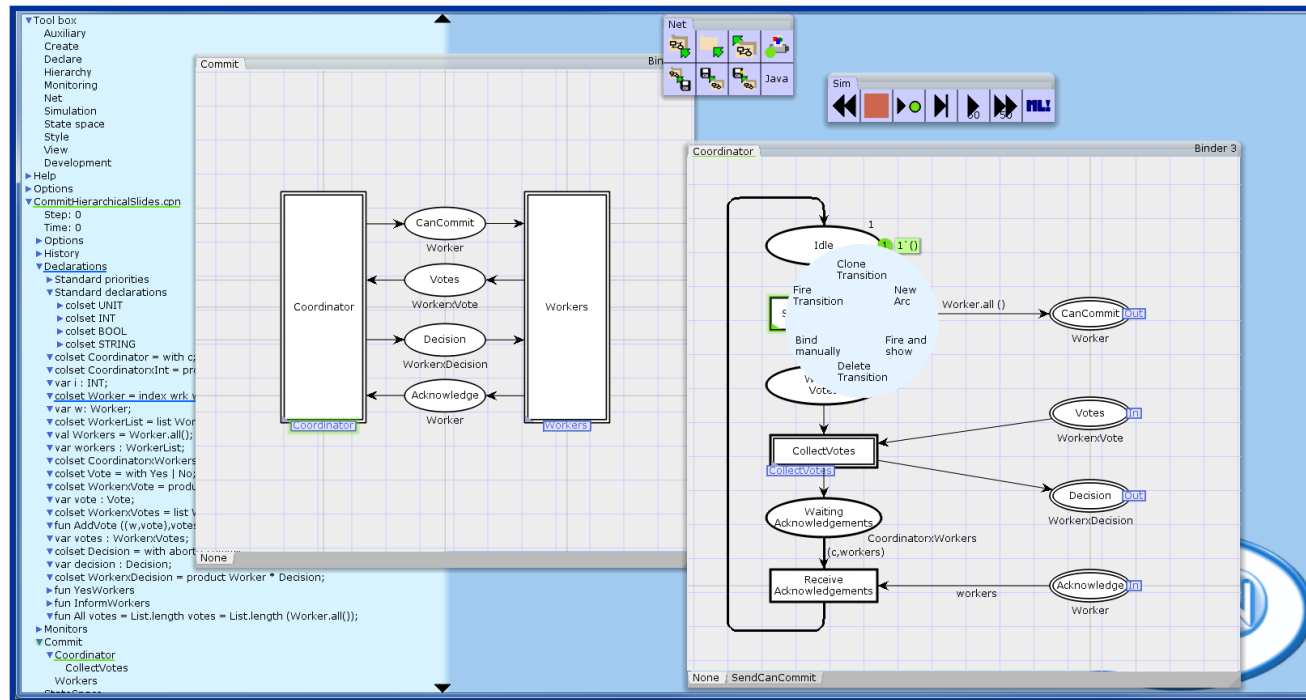
graphical notation
concurrency
communication
synchronisation
resource sharing

CPN ML (Standard ML)

data and data manipulation
compact modelling
parameterisable models

CPN Tools [www.cpntools.org]

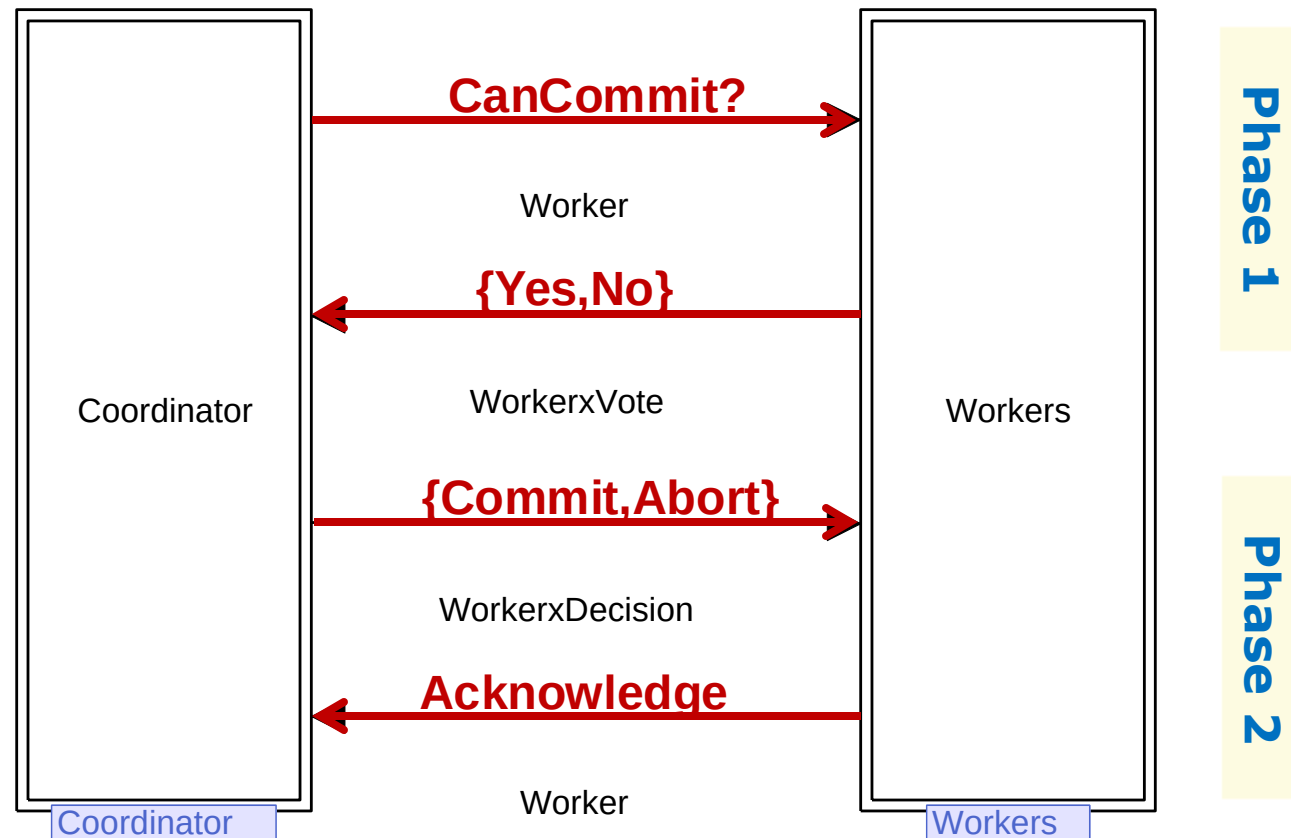
- Practical use of CPNs is supported by CPN Tools



- Editing and syntax check
- Interactive- and automatic simulation
- Verification based on state space exploration
- Simulation-based performance analysis

Example: Two-phase commit transaction protocol

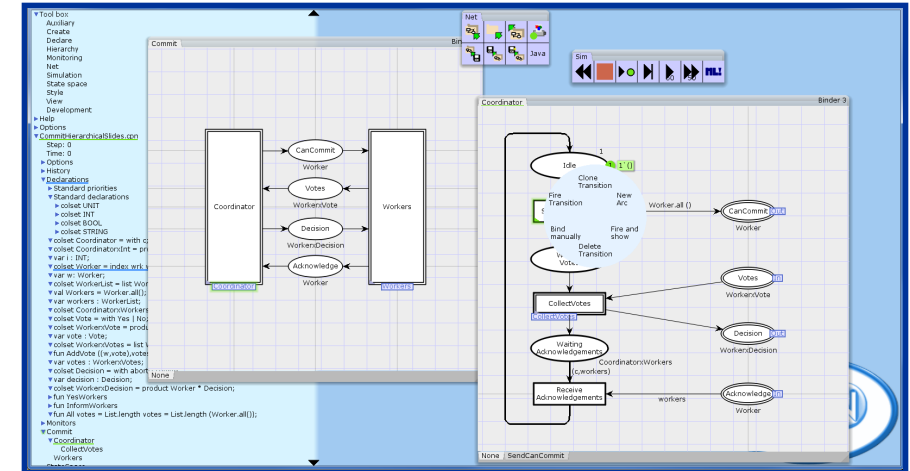
- A **concurrent system** consisting of a **coordinator process** and a number of **worker processes**



CPN Tools – initial demonstration

cpntools-demo.cpn

- **User-interaction with CPN Tools**
 - No menu-bars or dialog-boxes
 - Index and workspace
 - Binders and tool palettes
 - drag-and-drop
 - Contextual menus
 - right click



Examples of CPN Tools users

North America

- Boeing
- Hewlett-Packard
- Samsung Information Systems
- National Semiconductor Corp.
- Fujitsu Computer Products
- Honeywell Inc.
- MITRE Corp.,
- Scalable Server Division
- E.I. DuPont de Nemours Inc.
- Federal Reserve System
- Bell Canada
- Nortel Technologies, Canada

Asia

- Mitsubishi Electric Corp., Japan
- Toshiba Corp., Japan
- SHARP Corp., Japan
- Nippon Steel Corp., Japan
- Hongkong Telecom Interactive Multimedia System

Europe

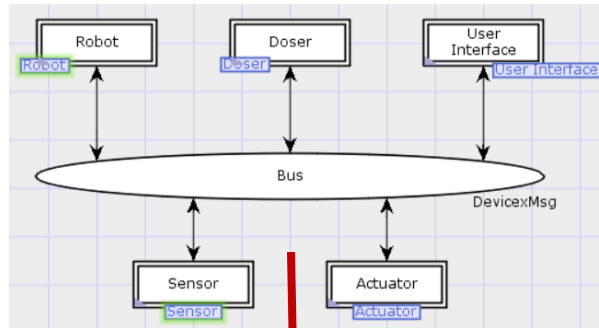
- Alcatel Austria
- Siemens Austria
- Bang & Olufsen, Denmark
- Nokia, Finland
- Alcatel Business Systems, France
- Peugeot-Citroën, France
- Dornier Satellitensysteme, Germany
- SAP AG, Germany
- Volkswagen AG, Germany
- Alcatel Telecom, Netherlands
- Rank Xerox, Netherlands
- Sydkraft Konsult, Sweden
- Central Bank of Russia
- Siemens Switzerland
- Goldman Sachs, UK

<http://cs.au.dk/cpnets/industrial-use/>

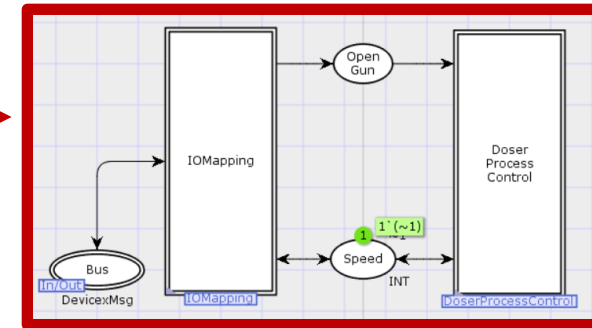
CPN @ Atlas Copco

- A model-driven software engineering approach for pump controllers of gluing robots in car assembly lines

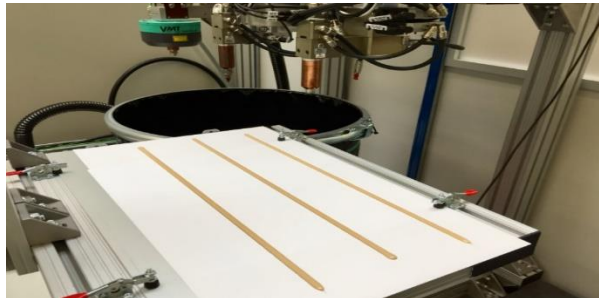
CPN Tools:
editing,
validation, and
verification
(design time)



Environment
modelling for
(non-site)
software testing

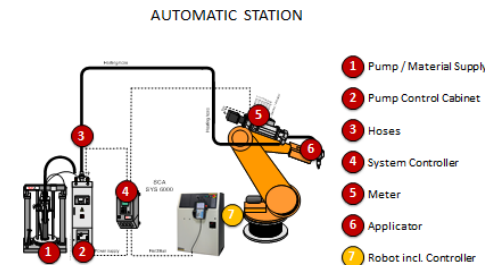


CPN execution
engine in C++ for
deployment and
real-time execution
(run-time)



SCA

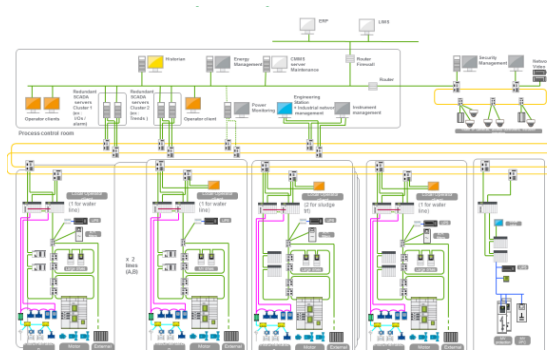
System Layouts



- The CPN model was **directly used** as implementation of the pump controller software

CPN @ Schneider Electric

- Dependability evaluation and capacity planning of large industrial automation architectures



Dependability analysis
software tools

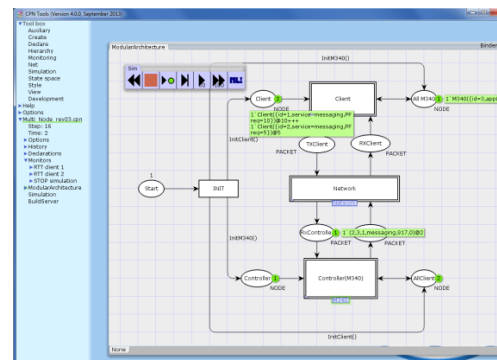


Performance - Reliability
Availability - Safety

Modelling



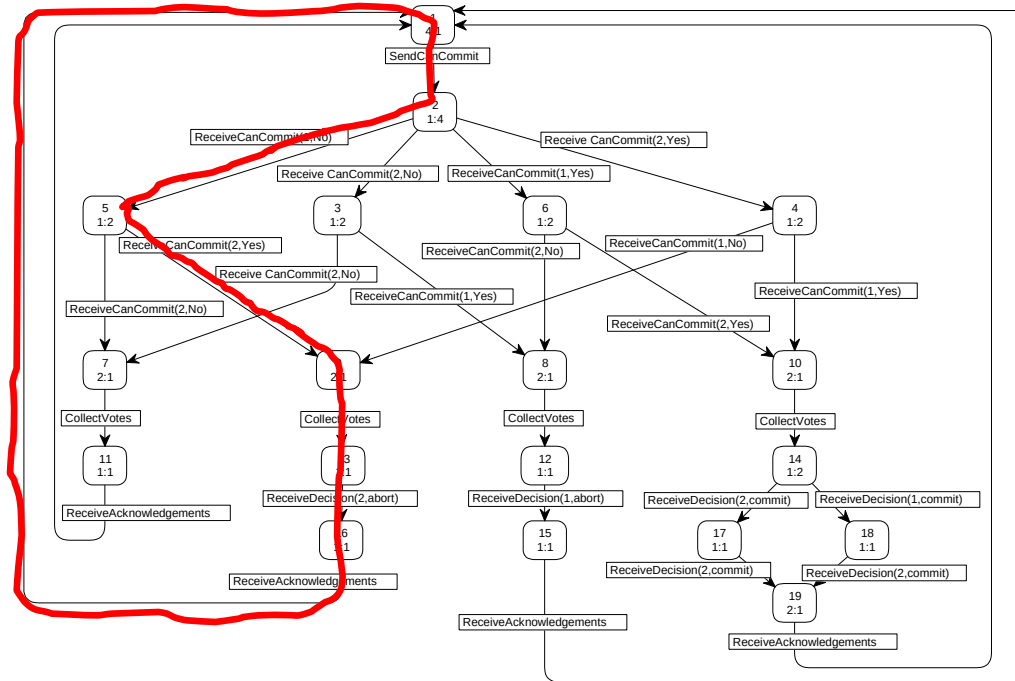
Automated
code generation



Tools for modelling driven
engineering

Verification and model checking

- **Formal verification** of CPN models can be conducted using **explicit state space exploration**

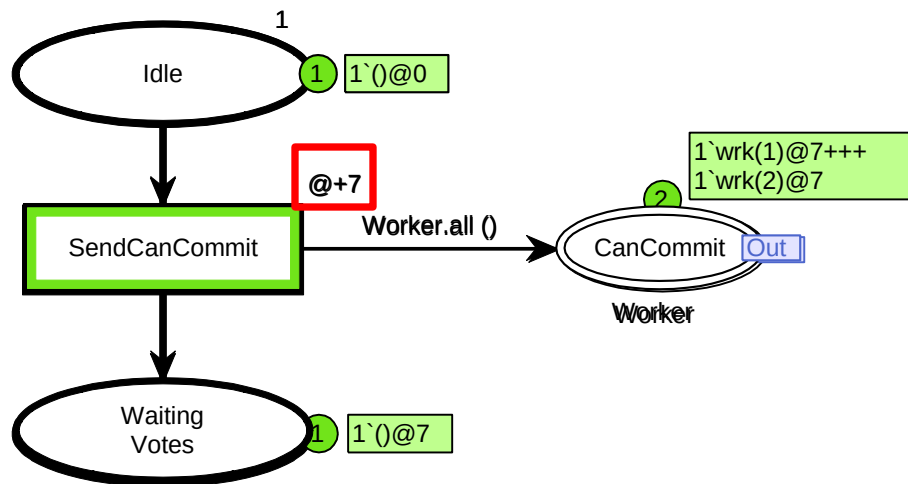


- A state space represents all possible **executions** of the CPN model
- **Standard behavioural properties** can be investigated using the state space report
- **Model-specific properties** can be verified using queries and **temporal logic model checking**

- Several **advanced techniques** available to alleviate the **inherent state explosion problem**

Performance analysis

- CPNs include a **concept of time** that can be used to model the time taken by activities



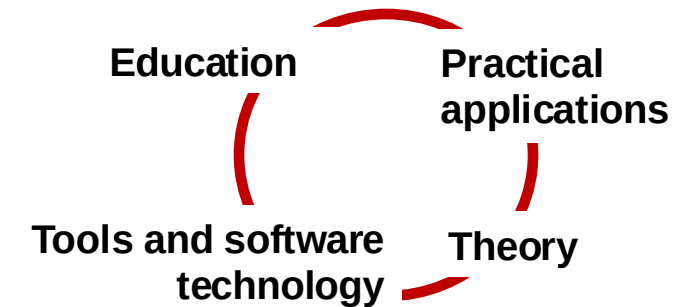
- A **global clock** representing the **current model time**
- Tokens carry **time stamps** describing the earliest possible model time at which they can be removed
- Time inscriptions** on transitions and arcs are used to give time stamps to the tokens produced on output places

- Random distribution functions** can be used in arc expressions (variable delays, packet loss probabilities, ...)
- Data collection monitors** and batch simulations can be used for **performance analysis**

Perspectives on CPNs



- **Modelling language combining Petri Nets with a programming language**
- **The development has been driven by an application-oriented research agenda**



- **Key characteristics**
 - Few but still powerful and expressive modelling constructs
 - **Implicit concurrency** inherited from Petri nets
 - everything is concurrent unless explicitly synchronised
 - **Verification** and **performance analysis** supported by the same modelling language